Deployed model results (app_model_results.json)

Contents of app_model_results.json — Medical Image Analysis Assistant

Selected model configuration (validation set)

Parameter	Value
config_id	70
lr	0.00100
batch_size	20
n_conv	4
n_dense	2
epochs	25
seconds	260.20000
val_accuracy	0.95763
val_sensitivity	0.95498
val_specificity	0.97674
val_precision	0.99664
val_f1	0.97537
val_mcc	0.83407
val_auc	0.98766

Test-set summary (four-class)

Metric	Value
Test images (n)	354
Trainable parameters	929,284
Epochs trained	25
Accuracy (95% CI)	0.7881 (0.7426–0.8275)
Macro F1	0.7895
Cohen's kappa	0.7098

Per-class performance (test set)

Class	n	Sensitivity (95% CI)	Specificity (95% CI)	AUC (OvR)
No tumour	43	0.884 (0.755–0.949)	0.952 (0.922–0.971)	0.963
Glioma	114	0.807 (0.725–0.869)	0.850 (0.799–0.890)	0.890
Meningioma	103	0.621 (0.525–0.709)	0.948 (0.913–0.969)	0.928
Pituitary tumour	94	0.904 (0.828–0.949)	0.958 (0.926–0.976)	0.980

Confusion matrix (rows = true class, columns = predicted class)

True \ Predicted	No tumour	Glioma	Meningioma	Pituitary tumour
No tumour	38	1	2	2
Glioma	10	92	10	2
Meningioma	4	28	64	7
Pituitary tumour	1	7	1	85

Binary collapse (tumour vs no tumour)

Metric	Value
TP	297
FN	14
FP	6
TN	37
N	354
Accuracy	0.9435
Sensitivity	0.9550
Specificity	0.8605
Precision	0.9802
NPV	0.7255
F1	0.9674
MCC	0.7586
AUC	0.9628

Raw data (original JSON)


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    "val_sensitivity": 0.95498,
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    "val_mcc": 0.83407,
    "val_auc": 0.98766
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      "spec_ci": [
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```